

Here are your notes on Bootstrapping, based on the two videos.

Lecture Notes: Bootstrapping

These two videos explain **bootstrapping**, a powerful computational method that uses a single sample to simulate what would happen if you repeated an experiment thousands of times. It's a way to understand the uncertainty of your statistics without complex formulas.

Part 1: Bootstrapping Main Ideas

1. The Problem: Uncertainty in a Small Sample

- **Scenario:** We give a new drug to 8 people. The average (mean) response is 0.5.
- **Question:** Is this 0.5 a real effect, or did we just get lucky with our 8 people? (e.g., maybe the 5 people who improved were healthier to begin with).
- **Traditional Solution (Expensive):** Repeat the *entire* 8-person experiment 1,000 times, get 1,000 means, and plot them on a histogram. This would show us the true variation, but it's impossibly expensive and time-consuming.

2. The Bootstrapping Solution: "Fake it 'til you make it"

Bootstrapping lets us *simulate* repeating the experiment using *only our original 8 data points*.

The 4-Step Bootstrapping Process:

1. Make a "Bootstrapped Dataset":

- Start with your original sample (of size $n = 8$).
- Create a *new* sample of the exact same size ($n = 8$) by randomly picking values from your original sample.
- The key is that you must use **sampling with replacement**. This means after picking a value, you "put it back," so you can (and often will) pick the same value multiple times.

2. Calculate the Statistic:

- Calculate your statistic of interest for this single bootstrapped dataset.
- For this example, we calculate the **mean** (which might be, say, 0.25).

3. Keep Track:

- Save this new mean.

4. Repeat:

- Repeat steps 1-3 thousands of times (e.g., 10,000 times).
- This gives you a large distribution (a histogram) of 10,000 bootstrapped means.

3. What This Bootstrapped Distribution Tells Us

This new histogram of bootstrapped means is a powerful proxy for the "expensive solution" histogram. We can now use it to measure uncertainty:

- **Standard Error (SE):** The **Standard Error of the Mean (SEM)** is simply the **standard deviation of this new bootstrapped distribution**. You get it directly from your results, no formula needed.
- **95% Confidence Interval (CI):** This is the interval that covers the central 95% of your 10,000 bootstrapped means.

4. The Power of Bootstrapping (Why it's "Awesome")

The real magic is that this works for *any statistic*, not just the mean.

- If you're worried about outliers, you can use the **median** instead.
- The "standard error of the median" has no simple formula.
- With bootstrapping, you don't need one! You just follow the 4-step process, but in Step 2, you calculate the **median** of each bootstrapped sample.
- The result is a bootstrapped distribution of *medians*, which you can use to find the SE and 95% CI for the median.

Part 2: Using Bootstrapping to Calculate p-values

Now we can use this idea to perform hypothesis testing.

- **Goal:** Calculate a p-value for our original observed mean of 0.5.
- **Null Hypothesis (H_0):** The drug has no effect (the true mean $\mu = 0$).
- **p-value Definition:** The probability of getting a result *at least as extreme* as our observed one (0.5), *if the null hypothesis were true*.

The distribution we made in Part 1 is centered at 0.5. To get a p-value, we need a new distribution that is centered at 0 (to represent the H_0).

The 3-Step Process for a Bootstrapped p-value

1. Create a Null Distribution:

- Take your original data (which has a mean of 0.5).
- **Shift the entire dataset** so that its mean is 0. We do this by subtracting 0.5 from *every single data point*.
- This new, shifted dataset now represents a "null world" where the drug has no effect, but the data has the same internal variation as our original sample.

2. Bootstrap the Null Distribution:

- Perform the 4-step bootstrap process (from Part 1) on this **new, shifted data**.
- The result is a histogram of thousands of bootstrapped means, all centered around 0. This shows us the range of means we'd expect to get *just from random chance* if the drug had no effect.

3. Calculate the p-value:

- Now, we see how "surprising" our original mean (0.5) is in this "null world."
- We find the proportion of bootstrapped means in the null distribution that are "at least as extreme" as 0.5.
- For a two-sided test, this means:
 - (Proportion of values ≥ 0.5)
 - **PLUS**
 - (Proportion of values ≤ -0.5)
- **Example from video:** $p\text{-value} = 0.47 + 0.16 = 0.63$.

Conclusion

- Our p-value (0.63) is much larger than 0.05.
- Therefore, we **fail to reject the null hypothesis**.
- This matches our conclusion from Part 1: We do not have strong statistical evidence that the drug works.
- This "shift-then-bootstrap" method also works for *any statistic* (like the median) for which you want to calculate a p-value.

http://googleusercontent.com/youtube_content/6 http://googleusercontent.com/youtube_content/7